Bounds on 21st-century WAIS mass loss from grounding-line flux theory

Technical note

1 Introduction

The West Antarctic Ice Sheet (WAIS) is the largest source of uncertainty in sea-level projections at centennial timescales. Marine ice sheet instability (MISI) on retrograde beds can drive self-sustaining grounding-line retreat, but the rate and magnitude of this retreat are constrained by ice dynamics and bed geometry. Here we derive two complementary bounds on the WAIS contribution to sea-level rise (SLR) during the 21st century, using the grounding-line flux laws of Schoof (2007), Pegler (2018), and Haseloff and Sergienko (2018).

1. A **geometric lower bound** ($\sim$200 mm SLE for WAIS): the minimum SLR from grounding-line retreat through the retrograde bed sections, independent of ice rheology (n), basal sliding (m), and the rate factor (A). If MISI is triggered, the bed geometry alone guarantees at least this much mass loss, regardless of the uncertain details of ice dynamics.
2. A **buttressing-dependent upper bound**: the maximum SLR achievable under a given level of residual buttressing. In the zero-buttressing limit, the dynamics are so fast ($s > 4.5$, finite-time singularity within decades) that the upper bound saturates at the total ice volume above flotation ($V_{\mathrm{af}} \sim 3.3$ m for WAIS). A non-trivial upper bound requires partial buttressing, parameterized by the framework-specific buttressing exponents of Pegler (2018) and Haseloff and Sergienko (2018).

The derivation connects the reduced dynamics of grounding-line flux theory to the scenario ranges of the A4 framework, providing physical grounding for both the lower and upper ends of the MISI scenario (S2: 150–1000 mm).

2 Grounding-line flux law

2.1 The Schoof flux condition

Schoof (2007) derived the grounding-line flux condition for a marine ice sheet using matched asymptotic expansions. For a power-law sliding law

$$\tau_b = C\,|u_b|^{m-1}\,u_b, \tag{1}$$

where m is the velocity exponent ($0 < m \leq 1$; $m \to 0$ for a perfectly plastic bed, $m = 1/n$ for Weertman sliding at Glen's law exponent n), the unbuttressed grounding-line flux is

$$q_g^{\mathrm{unbutt}} \propto h_g^{s_{\mathrm{full}}}, \qquad s_{\mathrm{full}} = \frac{m+n+3}{m+1}. \tag{2}$$

Two limiting cases confirm the physical content: for a perfectly plastic bed ($m \to 0$), $s_{\text{full}} \to n+3$—the flux is highly sensitive to thickness because ice viscosity provides the only resistance; for linear sliding ($m = 1$), $s_{\text{full}} = (n+4)/2$—basal drag provides significant resistance. In both limits $s_{\text{full}} > 2$ for $n \geq 3$.

2.2 Buttressing and the three frameworks

Ice-shelf buttressing reduces the flux below its unbuttressed value. Parameterizing the back-stress through a buttressing ratio $\theta \in [0, 1]$ ($\theta = 0$: no buttressing; $\theta = 1$: full buttressing), the flux scales as

$$q_g(\theta) = q_g^{\text{unbutt}} (1 - \theta)^{s_b}, \tag{3}$$

where the *buttressing exponent* s_b depends on the framework:

$$s_b^{(S)} = \frac{n}{m+1} \qquad \text{(Schoof: unconfined)}, \tag{4}$$

$$s_b^{(P)} = \frac{n+1}{m+1} \qquad \text{(Pegler: confined, viscous lateral drag)}, \tag{5}$$

$$s_b^{(H)} = \frac{n+1}{m+1} + \frac{1}{n} \qquad \text{(Haseloff: resolved shear margins)}. \tag{6}$$

For Weertman sliding ($m = 1/n$), table 1 collects the numerical values. The buttressing exponents s_b control how sensitively the flux responds to partial loss of buttressing; they are distinct from the full flux-thickness exponent s_{full}, which governs the response to grounding-line deepening.

Table 1: Flux and buttressing exponents for Weertman sliding ($m = 1/n$). The full flux exponent s_{full} enters the retreat dynamics; the buttressing exponents s_b enter the buttressing-dependent bound.

	$n = 3$, $m = 1/3$	$n = 4$, $m = 1/4$	Plastic ($m \to 0$)
$s_{\text{full}} = (m+n+3)/(m+1)$	4.75	5.80	$n+3$
$s_b^{(S)} = n/(m+1)$	2.25	3.20	n
$s_b^{(P)} = (n+1)/(m+1)$	3.00	4.00	$n+1$
$s_b^{(H)} = (n+1)/(m+1) + 1/n$	3.33	4.25	$n+1+1/n$

3 Retreat dynamics and the geometric lower bound

3.1 Setup

Consider a marine-terminating glacier with grounding-line position $x_g(t)$ on a retrograde bed

$$b(x) = -(b_0 + \alpha_b\, x), \tag{7}$$

where $b_0 > 0$ is the current bed depth and $\alpha_b > 0$ is the retrograde slope. At the grounding line, ice is at flotation:

$$h_g(x_g) = \phi\,(b_0 + \alpha_b\, x_g) = h_0 + \phi\, \alpha_b\, x_g, \tag{8}$$

with $\phi = \rho_{\text{w}}/\rho_{\text{i}} \approx 1.12$ and $h_0 = \phi\, b_0$.

The grounding-line retreat rate is (Schoof, 2012)

$$\dot{x_g} = \frac{q_g - q_{\text{in}}}{h_g}, \tag{9}$$

where q_{in} is the upstream resupply flux. Setting $q_{\text{in}} = 0$ and $\dot{a} = 0$ (zero accumulation) gives the fastest possible retreat. In this limit, with $q_g = q_0 (h_g/h_0)^{s_{\text{full}}}$ calibrated to the current discharge:

$$\frac{\mathrm{d}\hat{h}}{\mathrm{d}\tau} = \hat{h}^{\,s_{\text{full}}-1}, \qquad \hat{h}(0) = 1, \tag{10}$$

where $\hat{h} = h_g/h_0$, $\tau = t/t_*$, and

$$t_* = \frac{h_0}{\phi\,\alpha_b\,u_0} \tag{11}$$

is the *geometric timescale* ($u_0 = q_0/h_0$ is the current grounding-line velocity).

3.2 Finite-time singularity

For $s_{\text{full}} > 2$ (all physically relevant cases, table 1):

$$\hat{h}(\tau) = \left[1 - (s_{\text{full}} - 2)\,\tau\right]^{-1/(s_{\text{full}}-2)}, \tag{12}$$

which diverges at

$$t_{\text{sing}} = \frac{t_*}{s_{\text{full}} - 2}. \tag{13}$$

For Thwaites Glacier (table 2; $t_* \approx 116\,\text{yr}$) with $n = 4$ ($s_{\text{full}} = 5.8$): $t_{\text{sing}} \approx 31\,\text{yr}$ from onset, i.e., $\sim$2041. This is not a prediction of actual behavior—it reflects the extreme zero-resupply, zero-buttressing assumptions—but it demonstrates that the flux dynamics are *fast* relative to the 21st-century timescale.

3.3 The geometric mass-loss identity

The total SLR from a marine ice sheet satisfies the mass conservation identity

$$H_{\text{SLR}}(T) = \frac{1}{\rho_{\text{w}}\,A_{\text{oc}}} \int_0^T \big(Q_g(t) - \dot{a}_{\text{total}}\big)\,\mathrm{d}t, \tag{14}$$

where $Q_g = \rho_{\text{i}}\,q_g\,W$ is the total grounding-line discharge and $\dot{a}_{\text{total}}$ is the total accumulation over the grounded domain. This identity holds for the full coupled system: sustaining grounding-line discharge Q_g requires interior drawdown that reduces the mass above flotation. Even though ice at the grounding line is at flotation (zero local mass above flotation), the process of sustaining the discharge thins the interior, and it is this interior thinning that raises sea level.

In the zero-resupply, zero-accumulation limit, the total mass crossing the grounding line during the traverse of the retrograde section from b_0 to $b_{\max}$ can be computed by changing variables from t to x_g. Since $q_g\,\mathrm{d}t = h_g\,\mathrm{d}x_g$ in this limit:

$$\int_0^{t_{\max}} q_g\,\mathrm{d}t = \int_0^{\Delta x} h_g\,\mathrm{d}x_g = \int_0^{\Delta x} \phi(b_0 + \alpha_b\,x)\,\mathrm{d}x = \frac{\phi}{2}\big(b_{\max}^2 - b_0^2\big)\,\frac{1}{\alpha_b}. \tag{15}$$

Proposition 1 (Geometric lower bound)**.** *If MISI drives the grounding line through the full retrograde section (bed depths b_0 to $b_{\max}$), the resulting SLR satisfies*

$$\boxed{H_{\mathrm{SLR}} \geq \frac{\rho_{\mathrm{i}}\, W\, \phi}{2\, \rho_{\mathrm{w}}\, A_{\mathrm{oc}}\, \alpha_b}\big(b_{\max}^2 - b_0^2\big) = \frac{Q_0\, t_*}{\rho_{\mathrm{w}}\, A_{\mathrm{oc}}} \cdot \frac{\hat{h}_{\max}^2 - 1}{2},} \tag{16}$$

where $\hat{h}_{\max} = \phi\, b_{\max}/h_0$. This bound is independent of n, m, the rate factor A, and the sliding coefficient C. It depends only on the bed profile (b_0, $b_{\max}$, α_b), glacier width (W), and the current state (h_0, Q_0).

The s-independence follows from the change of variables: $q_g\,\mathrm{d}t = h_g\,\mathrm{d}x_g$ eliminates the flux law entirely. All that s controls is *how fast* the traverse occurs, not *how much mass* is lost during it.

Remark: why this is a lower bound. The traverse integral counts only the discharge accumulated while the grounding line sweeps the retrograde section. After reaching maximum bed depth, the grounding line continues to discharge ice—potentially at very high rates—drawing down interior ice above flotation. This post-traverse contribution can be very large (see section 4). Moreover, in a fully coupled system with interior resupply, the total integrated discharge $\int Q_g\,\mathrm{d}t$ exceeds the traverse integral even during the retrograde phase, because interior ice flowing toward the grounding line is an additional source of discharge mass. The traverse integral is therefore a strict lower bound on the SLR from MISI.

3.4 Numerical evaluation

Table 2: Representative parameters for Thwaites Glacier and the Amundsen Sea Embayment (ASE). Bed depths and slopes from BedMachine Antarctica (Morlighem et al., 2020). Discharges from Rignot et al. (2019). The geometric timescale is $t_* = h_0/(\phi\,\alpha_b\,u_0)$.

Parameter	Thwaites	ASE (total)	Source
GL width, W	120 km	300 km	Remote sensing
Current bed depth, b_0	700 m	600–900 m	BedMachine
Flotation thickness, h_0	785 m	—	Derived
Max trough depth, $b_{\max}$	2000 m	1200–2000 m	BedMachine
$\hat{h}_{\max}$	2.85	1.5–2.9	Derived
Retrograde slope, α_b	0.0052	0.003–0.008	BedMachine
Total discharge, Q_0	100 Gt/yr	250 Gt/yr	Rignot et al. (2019)
GL velocity, u_0	1.2 km/yr	0.5–4 km/yr	InSAR
t_*	116 yr	80–200 yr	Eq. (11)
V_{af}	∼600 mm	∼1100 mm	Morlighem et al. (2020)

Table 3 evaluates the geometric lower bound for the major ASE glacier systems and for WAIS as a whole.

The geometric lower bound for all of WAIS is approximately **200 mm SLE**. This is the minimum SLR from MISI if the grounding lines traverse their full retrograde sections, regardless of the ice rheology. The traverse times (∼15–30 yr for individual ASE glaciers at $n = 4$) confirm that the full retrograde sections can be traversed well within the 21st century under the zero-resupply dynamics, so the bound is achievable within the projection horizon.

Table 3: Geometric lower bound on SLR from MISI retrograde retreat. These values are independent of n, m, and s. The traverse time $t_{\max}$ is shown for $n = 4$ ($s_{\text{full}} = 5.8$) to confirm that the full retrograde section can be traversed within the century under zero-resupply dynamics.

System	Lower bound (mm SLE)	$t_{\max}$ (n=4)	V_{af} (mm SLE)
Thwaites	114	∼30 yr	∼600
Pine Island	∼30	∼20 yr	∼400
Smith/Kohler	∼10	∼15 yr	∼100
ASE total	**∼155**	—	∼1100
Other WAIS	∼50	—	∼2200
Total WAIS	**∼200**	—	∼3300

4 Buttressing-dependent upper bound

4.1 The zero-buttressing ceiling: V_{af}

After the grounding line passes the deepest point of the retrograde section, discharge continues—potentially at very high rates. The peak flux ratio at maximum bed depth is $\hat{q}_{\max} = \hat{h}_{\max}^{s_{\text{full}}}$. For Thwaites with $\hat{h}_{\max} = 2.85$ and $s_{\text{full}} = 5.8$: $\hat{q}_{\max} \approx 390$. At 390 times the current discharge (∼39,000 Gt/yr), the entire Thwaites ice volume above flotation (∼217,000 Gt) would be drained in under 6 years.

This post-traverse flux is physically unrealizable at face value—it would require ice velocities ∼100× current values—but it establishes that in the zero-buttressing limit, the flux dynamics are fast enough to mobilize the *entire* ice volume above flotation within the 21st century. The upper bound in this limit is simply

$$H_{\text{SLR}}^{\max} = V_{\text{af}}. \tag{17}$$

For WAIS, $V_{\text{af}} \approx 3.3\,\text{m SLE}$—a trivially large bound that provides no useful constraint.

4.2 Non-trivial bound with partial buttressing

A useful upper bound requires residual buttressing ($\theta > 0$). With buttressing, the flux is reduced by a factor $(1-\theta)^{s_b}$ relative to the unbuttressed case, (3). This has two effects:

1. The *instantaneous discharge rate* is reduced, slowing the drawdown of interior ice.
2. The *retreat dynamics* are slowed, because the flux-thickness feedback is weaker when buttressing absorbs part of the driving stress.

For a constant buttressing ratio θ, the retreat ODE becomes

$$\frac{\mathrm{d}\hat{h}}{\mathrm{d}\tau} = (1-\theta)^{s_b}\,\hat{h}^{\,s_{\text{full}}-1}, \qquad \hat{h}(0) = 1, \tag{18}$$

which has the same analytical solution as (12) but with an effective timescale $t_*^{\text{eff}} = t_*/(1-\theta)^{s_b}$. The singularity time becomes

$$t_{\text{sing}}^{\text{eff}} = \frac{t_*}{(s_{\text{full}}-2)\,(1-\theta)^{s_b}}. \tag{19}$$

Table 4: Effect of buttressing on Thwaites dynamics ($n = 4$, Pegler framework). θ: buttressing ratio. $t_{\rm sing}^{\rm eff}$: time to mathematical singularity. $\hat{q}(T)$: discharge amplification at 2100. $M(T)$: cumulative discharge (mm SLE), capped at $V_{\rm af} = 600$ mm.

θ	$(1-\theta)^{s_b}$	$t_{\rm sing}^{\rm eff}$ (yr)	$\hat{q}(T=90)$	M (mm SLE)
0	1.000	31	$\gg 1$	600 ($= V_{\rm af}$)
0.1	0.656	47	$\gg 1$	600 ($= V_{\rm af}$)
0.2	0.410	75	$\gg 1$	600 ($= V_{\rm af}$)
0.3	0.240	127	129	$\sim$570
0.4	0.130	235	10.6	$\sim$300
0.5	0.063	490	3.1	$\sim$160
0.6	0.026	1180	1.6	$\sim$100
0.7	0.008	3700	1.1	$\sim$70

Even modest buttressing dramatically extends the singularity time. For Thwaites ($t_* = 116$ yr, $s_{\rm full} = 5.8$) with Pegler buttressing ($s_b^{(P)} = 4.0$, appropriate for confined ASE glaciers):

The table reveals a sharp transition: for $\theta \lesssim 0.3$, buttressing is insufficient to prevent the singularity from being reached within the century, and the upper bound saturates at $V_{\rm af}$. For $\theta \gtrsim 0.4$, the dynamics are slow enough that only a fraction of $V_{\rm af}$ is mobilized, yielding a non-trivial upper bound.

The cumulative discharge at year T, accounting for the bed-geometry truncation, is

$$M(T) = \min\left(V_{\rm af},\ \ Q_0\, t_*^{\rm eff} \left[\frac{\hat{h}(T)^2 - 1}{2} + \hat{h}_{\max}^{s_{\rm full}} \cdot \left(\hat{T}^{\rm eff} - \tau_{\max}\right)^+ \right] \right), \tag{20}$$

where $\hat{T}^{\rm eff} = T/t_*^{\rm eff}$ and $\hat{h}(T)$ is evaluated from (12) at $\tau = T/t_*^{\rm eff}$ (or $\hat{h}_{\max}$ if the traverse completes before T).

4.3 Buttressing loss trajectory

In reality, θ is not constant: buttressing decreases as ice shelves thin and retreat. A time-varying $\theta(t)$ can be incorporated by solving (18) numerically, or by specifying a buttressing-loss timescale τ_θ and parameterizing $\theta(t) = \theta_0\, e^{-t/\tau_\theta}$ (exponential loss from initial value θ_0). For rapid shelf collapse ($\tau_\theta \sim 10$ yr), the system transitions quickly from the buttressed to the unbuttressed regime, and the upper bound approaches $V_{\rm af}$. For gradual thinning ($\tau_\theta \sim 100$ yr), significant buttressing persists throughout the century, keeping the upper bound well below $V_{\rm af}$.

Naughten et al. (2023) showed that West Antarctic ice-shelf basal melt rates of $\sim 3\times$ historical are unavoidable regardless of emissions scenario, implying $\tau_\theta \sim 50$–100 yr for melt-driven buttressing loss. Joughin et al. (2014) documented ongoing retreat of the Thwaites grounding line on a retrograde bed, suggesting the transition from buttressed to partially unbuttressed dynamics is already underway.

5 Connection to the A4 scenario framework

5.1 Mapping scenarios to buttressing states

The A4 scenario framework defines three WAIS regimes: S1 (status quo, 30–80 mm), S2 (MISI, 150–1000 mm), and S3 (MISI + MICI, 600–2000 mm). The bounds derived here provide physical interpretation for these ranges:

S1 (30–80 mm). No MISI: the grounding line does not traverse the retrograde section. Discharge continues at roughly current rates ($\sim$0.4 mm/yr) without instability amplification. This corresponds to θ remaining large enough to stabilize the grounding line.

S2 lower bound ($\sim$150 mm). Coincides with the geometric lower bound for the ASE ($\sim$155 mm, table 3). If MISI is triggered and the grounding lines traverse their retrograde sections, the bed geometry alone guarantees at least this much SLR. The agreement between the S2 lower bound and the geometric bound is not coincidental: it reflects the fact that 150 mm is roughly the minimum dynamically consistent with MISI across the ASE.

S2 range (150–1000 mm). The full S2 range corresponds to buttressing ratios of $\theta \approx 0.7$ (lower end, $\sim$70 mm for Thwaites) to $\theta \approx 0.2$ (upper end, approaching V_{af} for individual glaciers). The S2 upper bound of 1000 mm requires either low residual buttressing ($\theta \lesssim 0.3$) across most ASE basins, or amplifying feedbacks (calving, subglacial hydrology (Dow et al., 2022)) that effectively reduce θ faster. The positive skewness ($\alpha = +4$) assigned to S2 is consistent with the sharp nonlinearity in table 4: a modest reduction in buttressing (from $\theta = 0.5$ to $\theta = 0.3$) more than triples the mass loss.

S3 (600–2000 mm). Exceeds the Thwaites V_{af} of $\sim$600 mm and approaches the ASE V_{af} of $\sim$1100 mm. Reaching these values requires either near-complete deglaciation of multiple ASE basins or an additional mass-loss mechanism—marine ice cliff instability (MICI)—that bypasses the grounding-line flux law (Bassis and Walker, 2012; DeConto and Pollard, 2016). Recent evidence suggests MICI is unlikely to dominate during the 21st century (Morlighem et al., 2024; Clerc et al., 2019; Schlemm et al., 2022), consistent with the low weight ($P = 0.10$) and negative skewness ($\alpha = -3$) assigned to S3.

5.2 The trajectory exponent β

The A4 framework parameterizes temporal structure through $H(t) = H_{2100}\,(t/T)^{\beta}$. The connection to buttressing is:

- High θ (strong buttressing) $\Rightarrow$ slow retreat, nearly constant discharge $\Rightarrow$ $\beta \approx 1$ (linear).
- Low θ (weak buttressing) $\Rightarrow$ fast retreat, accelerating discharge $\Rightarrow$ $\beta > 1$ (back-loaded).
- Decreasing $\theta(t)$ (progressive shelf loss) $\Rightarrow$ increasing flux nonlinearity through the century $\Rightarrow$ strongly back-loaded ($\beta \gg 1$).

The A4 prior on β for S2 (LN(ln 1.8, 0.3), median 1.8) is consistent with gradual buttressing loss over the century, transitioning from a buttressed regime ($\beta \approx 1$) in the early decades to a weakly buttressed regime ($\beta > 2$) later.

The buttressing exponents s_b enter through the rheology correction: when correcting from $n_{\mathrm{ref}} = 3$ to $n = 4$, the trajectory exponent scales as $\beta(n)/\beta(n_{\mathrm{ref}}) = s_b(n)/s_b(n_{\mathrm{ref}})$. For the Pegler framework: $\beta(n=4)/\beta(n=3) = 4.0/3.0 = 1.33$; correcting $\beta = 1.8$ at $n = 3$ gives $\beta \approx 2.4$ at $n = 4$.

6 Discussion

6.1 Complementarity of the two bounds

The geometric lower bound and the buttressing-dependent upper bound bracket the MISI contribution from opposite directions:

- The **lower bound** ($\sim$200 mm) answers: *what is the minimum SLR from MISI?* It is rheology-independent and depends only on what we know well (bed geometry, current discharge). Its weakness is that it only counts mass from the retrograde traverse, ignoring the potentially larger contribution from interior drawdown.
- The **upper bound** (V_{af} at zero buttressing; non-trivial with partial buttressing) answers: *what is the maximum SLR given residual buttressing?* It accounts for the full interior ice reservoir but depends on the buttressing ratio θ and the framework-specific exponent s_b, both of which are uncertain.

Together, they establish that the S2 range (150–1000 mm) is physically well-motivated: the lower end is set by bed geometry (achievable regardless of rheological uncertainty), and the upper end is set by the rate of buttressing loss (requiring near-complete shelf collapse across most ASE basins).

6.2 Comparison with Pfeffer et al. (2008)

Pfeffer et al. (2008) bounded glacier contributions to SLR by prescribing velocity amplification factors. Our approach derives the amplification from the flux law, distinguishing between the full thickness exponent s_{full} (which controls retreat speed) and the buttressing exponent s_b (which controls sensitivity to shelf loss). The geometric lower bound is more robust than Pfeffer's kinematic bound because it is independent of the assumed velocity amplification.

6.3 Limitations

1. **Linear bed.** Real beds have local ridges and variable slope (Morlighem et al., 2020). The lower bound should be refined by integrating over actual BedMachine profiles: $M_{\mathrm{traverse}} = \rho_{\mathrm{i}} W \int_{x_0}^{x_{\max}} \phi\, |b(x)|\, \mathrm{d}x$.
2. **Constant θ.** Buttressing evolves with ice-shelf geometry. A realistic upper bound requires coupling $\theta(t)$ to an ice-shelf evolution model, or parameterizing the buttressing-loss trajectory.
3. **Width variation.** The Thwaites trough broadens inland from $\sim$120 km to $\sim$200 km. Width variation enters the lower bound through $W(x)$ in the traverse integral.
4. **Steady-state flux law.** During rapid retreat, transient effects may cause the flux to deviate from the steady-state boundary-layer scaling (Schoof, 2012; Haseloff and Sergienko, 2022). The steady-state flux law approximates the transient behavior well when the retreat rate is small compared to the ice velocity, a condition that breaks down near the singularity.

Summary

The grounding-line flux law provides two complementary bounds on 21st-century WAIS mass loss from MISI:

1. A **geometric lower bound** of $\sim$200 mm SLE, equal to the mass discharged during grounding-line retreat through the full retrograde bed sections of all WAIS basins. This bound is independent of the Glen's law exponent n, the sliding law exponent m, the rate factor A, and the sliding coefficient C. It depends only on the bed geometry, glacier width, and current discharge—quantities known from BedMachine and satellite observations. If MISI is triggered, this is the minimum guaranteed SLR.

2. A **buttressing-dependent upper bound** that transitions from $\sim$100 mm (60–70% buttressing retained) through $\sim$300 mm ($\sim$40% retained) to the full ice volume $V_{\mathrm{af}} \approx 3.3$ m ($< 30\%$ retained). The transition is sharp: a reduction in buttressing from 50% to 30% more than triples the projected mass loss, reflecting the high buttressing exponents ($s_b \sim 3$–4) predicted by the Pegler and Haseloff frameworks.

The A4 S2 scenario range (150–1000 mm) spans from just below the geometric lower bound to the regime of substantial interior drawdown under low buttressing. The lower end is guaranteed by bed geometry; the upper end requires near-complete shelf loss across most ASE basins. Both bounds arise from the same flux-law physics, applied at opposite ends of the buttressing spectrum.

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